



Darshan
UNIVERSITY

Python Programming - 2301CS404

Lab - 4

Roll No.: 459

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01) WAP to print 1 to 10.

```
In [2]: for i in range(1,11):  
        print(i)
```

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10
```

02) WAP to print 1 to n.

```
In [3]: n = int(input("Enter Number n : "))  
  
        for i in range(1,n+1):  
            print(i)
```

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18

03) WAP to print odd numbers between 1 to n.

```
In [4]: n = int(input("Enter Number n : "))  
  
for i in range(3,n,2):  
    print(i)
```

1
3
5
7
9
11
13

04) WAP to print numbers between two given numbers which is divisible by 2 but not divisible by 3.

```
In [6]: num1 = int(input("Enter Starting Number : "))  
num2 = int(input("Enter Ending Number : "))  
  
for i in range(num1+1 , num2):  
    if(i % 2 == 0 and i % 3 != 0):  
        print(i)
```

2
4
8
10
14
16

05) WAP to print sum of 1 to n numbers.

```
In [10]: n = int(input("Enter Number : "))  
  
sum = 0  
  
for i in range(1,n+1):
```

```
sum += i

print(sum)
```

28

06) WAP to print sum of series $1 + 4 + 9 + 16 + 25 + 36 + \dots n$.

```
In [12]: n = int(input("Enter Number : "))

sum = 0

for i in range(1,n+1):
    sum += i*i

print(sum)
```

140

07) WAP to print sum of series $1 - 2 + 3 - 4 + 5 - 6 + 7 \dots n$.

```
In [14]: n = int(input("Enter Number : "))

sum = 0

# for i in range(1,n+1):
#     if(i % 2 == 0):
#         sum -= i
#     else:
#         sum += i

sign = 1

for i in range(1,n+1):
    sum += i * sign

    sign *= -1

print(sum)
```

2

08) WAP to print Multiplication Table of the given number.

```
In [17]: n = int(input("Enter Number : "))

for i in range(1,11):
    print(n , "*" , i , "=" , n * i)
```

```

19 * 1 = 19
19 * 2 = 38
19 * 3 = 57
19 * 4 = 76
19 * 5 = 95
19 * 6 = 114
19 * 7 = 133
19 * 8 = 152
19 * 9 = 171
19 * 10 = 190

```

09) WAP to find Factorial of the given number.

```

In [18]: n = int(input("Enter Number : "))

fac = 1

for i in range(n,0,-1):
    fac *= i

print(fac)

```

120

10) WAP to print GCD of given two numbers.

```

In [6]: a = int(input("Enter Number 1 : "))
b = int(input("Enter Number 2 : "))

# while(b != 0):
#     a,b = b,a%b
# print(a)

min_n = min(a,b)

for i in range(min_n,0,-1):
    if(b % i == 0):
        print(i)
        break

```

15

11) WAP to find Factors of the given number.

```

In [9]: n = int(input("Enter Number : "))

for i in range(1,n+1):
    if(n % i == 0):
        print(i,end = " ")

```

1 2 3 6

12) WAP to find whether the given number is Prime or not.

```

In [14]: n = int(input("Enter Number : "))

for i in range(2,int(pow(n,1/2))+1): # ahiya java ni jem sqrt function nai ave s

```

```

        if(n % i == 0):
            print("No",n,"Is Not Prime Number")
            break
    else:
        print("Yes",n,"Is Prime Number")

```

No 18 Is Not Prime Number

13) WAP to print sum of digits of given number.

```

In [17]: n = int(input("Enter Number : "))

sum = 0

while(n != 0):
    sum += n%10
    n //= 10

print(sum)

```

6

14) WAP to check whether the given number is Palindrome or not.

```

In [26]: n = int(input("Enter Number : "))
original_num = n
# n2 = n[::-1]

# if(n == n2):
#     print(n,"Is Palindrome")
# else:
#     print(n,"Is Not Palindrome")
#     print(n,"Is Not Palindrome")

#or

rev = 0

while n != 0:
    rev = rev * 10 + (n % 10)
    # rev*=10+ (n % 10)=> rev=10( rev+(n % 10))
    n //= 10

if(rev == original_num):
    print(original_num,"Is Palindrome")
else:
    print(original_num,"Is Not Palindrome")

```

353 Is Palindrome

15) WAP to check whether the given number is an Armstrong Number or not.

```

In [28]: n = input("Enter Number : ")

power = len(n)

```

```

n = int(n)

original_num = n

sum = 0

while(n != 0):
    sum += (n%10) ** power
    n //= 10

if(sum == original_num):
    print(original_num,"Is Armstrong")
else:
    print(original_num,"Is Not Armstrong")

```

153 Is Armstrong

16) WAP to print all the perfect numbers between 1 to n.

In [34]:

```

n = int(input("Enter Number : "))

for i in range(2,n):
    sum = 0
    for j in range(1,i):
        if(i % j == 0):
            sum += j
    else:
        if(sum == i):
            print("Perfect Num : ",i)

```

Perfect Num : 6

Perfect Num : 28